

## CSC-324: Computer Organization and Assembly Language

### General Information

|                          |                                                       |
|--------------------------|-------------------------------------------------------|
| <b>Course Number</b>     | CSC-324                                               |
| <b>Credit Hours</b>      | 3(2+1) (Theory Credit Hour = 3, Lab Credit Hours = 1) |
| <b>Prerequisite</b>      | Digital Logic Design                                  |
| <b>Course Instructor</b> | Bilal Ahmed                                           |

### Course Objectives

This course introduces the organization of digital computer systems, focusing on how hardware components and assembly-level programming interact. Major topics include the organization of processor, memory, and input/output modules, as well as the interconnections among these components. The course examines the central processing unit and its functional units, including the role of registers and their organization. Students will gain an understanding of instruction formats and the basic instruction sets of 16-bit and 32-bit processors. Emphasis is placed on developing skills in assembly language programming, including program structure, instruction usage, and programming techniques, to bridge theoretical organization concepts with practical implementation.

### Catalog Description

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| CSC 324 |
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| Week(s) | Topic                                | Subtopics                                                                                                             | Suggested Reading   | CLO |
|---------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------|-----|
| 1       | Introduction to the Course & Subject | 1.1 Organization and Architecture<br>1.2 Structure & Function<br>1.3 Why study computer organization and architecture | Book [1], Chapter 1 | 1   |
| 2–3     | Computer Evolution and Performance   | 2.1 A brief history of computers<br>2.2 Designing for performance<br>2.3 Overview of processor evolution              | Book [1], Chapter 2 | 1   |

|       |                                        |                                                                                                                                                          |                                             |   |
|-------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---|
| 4–5   | Computer Function and Interconnection  | 3.1 Computer components 3.2 Computer functions 3.3 Interconnection structures 3.4 Bus interconnection                                                    | Book [1], Chapter 3                         | 1 |
| 6     | Memory Organization                    | 4.1 Overview of memory hierarchy 4.2 Cache memory principles (basic concepts) 4.3 Brief introduction to main memory and external storage                 | Book [1], Chapters 4–5 (selected sections)  | 1 |
| 7     | Input / Output Organization            | 5.1 External devices 5.2 I/O modules 5.3 Programmed I/O 5.4 Interrupt-driven I/O (basic introduction)                                                    | Book [1], Chapter 7 (selected sections)     | 2 |
| 8     | Number Systems and Data Representation | 6.1 Decimal Number System 6.2 Binary Number System 6.3 Octal Number System 6.4 Hexadecimal Number System 6.5 Signed and unsigned numbers 6.6 Complements | Book [1], Chapter 9                         | 2 |
| 9     | Computer Arithmetic (Basic Concepts)   | 7.1 Binary addition and subtraction 7.2 Multiplication and division (introductory level) 7.3 Floating-point representation (overview only)               | Book [1], Chapters 9–10 (selected sections) | 2 |
| 10–11 | Processor Structure and Function       | 8.1 Processor organization 8.2 Register organization 8.3 Instruction cycle 8.4 Introduction to instruction pipelining                                    | Book [1], Chapter 14 (selected sections)    | 2 |
| 12    | Course Review and Discussion           | Recap of major concepts<br>Preparation for final examination                                                                                             | –                                           | – |

### Text Book

1. Computer Organization and Architecture By: William Stallings Prentice Hall
2. Ytha YU and Charles Marut, “Assembly Language Programming and Organization of the IBM PC”

## Reference Material

1. Kip R Irvine, "Assembly Language for Intel-based Computers"
2. Essentials of Computer Architecture By: Douglas E. Comer

## Course Learning Outcomes

|   | <b>Course Learning Outcomes (CLO)</b>                                                                                                          |
|---|------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Understanding the nature and characteristics of modern computer systems architecture with employed Low level programming tools and techniques. |
| 2 | To Apply low-level programming logic employed for problem solving with respect to organization of computer system.                             |
| 3 | To Analyze the problem and mapping in the different nature of instruction set models.                                                          |

## CLO-SO Map

[illegible]